

Problem Statement and Research Gap

1. The Current Landscape Recent advancements in computational disaster management have primarily concentrated on discriminative models, successfully executing vision-based classification tasks. For instance, frameworks like BanglaMM-Disaster have established robust regional benchmarks for categorical severity assessment in the Bengali language. Concurrently, studies such as BiMCF have validated the efficacy of multimodal cascaded fusion. Parallel to discriminative classification, generative models like BiCap have introduced Bengali image captioning within general domains, while architectures like Flood-XAI demonstrated the critical value of explainable generative disaster reporting, albeit strictly confined to the high-resource English linguistic context.

2. The Core Limitations Despite these isolated successes, a critical architectural void persists: the absolute absence of a unified Vision-Language Model (VLM) capable of executing simultaneous classification and generative reporting for low-resource disaster environments. Existing multimodal frameworks remain fundamentally discriminative, intrinsically incapable of producing explainable descriptive narratives. Conversely, generative attempts in Bengali (e.g., BiCap) rely on unoptimized, outdated LSTM architectures unsuitable for catastrophic visual semantics. Furthermore, seemingly comprehensive systems like Flood-XAI rely on disjointed pipelines and completely separate datasets for their subtasks. Crucially, none of these baseline frameworks explicitly address the mathematical challenge of negative transfer and gradient interference inherent in multi-task joint training.

3. Our Novel Contribution: DisasterNet-Bangla To bridge this multifactorial gap, this research proposes DisasterNet-Bangla, a novel end-to-end Multi-Task Vision-Language framework. By completely transitioning from outdated CNNs to a robust Shared Vision Transformer (ViT) encoder and utilizing a LoRA-fine-tuned BanglaBERT decoder, the system executes simultaneous real-time flood severity classification and context-aware Bengali caption generation in a single forward pass. Crucially, to mitigate catastrophic forgetting and task interference between the data-rich classification head and the data-scarce generative head, we explicitly inject task-routing tokens (e.g., <CLASSIFY>, <CAPTION>) during the shared encoding phase. This explicit Multi-Task Learning (MTL) architecture explicitly models cross-modal interdependencies, maximizing computational efficiency while providing an explainable, dual-output AI system tailored for rapid humanitarian response.

Research Objectives (RO)

The primary objective of this research is to develop an efficient and real-time Artificial Intelligence (AI) framework for disaster management, specifically addressing the challenges of data scarcity and linguistic barriers in low-resource environments like Bangladesh. The specific objectives are:

- **RO1 (Architecture Development):** To design and develop a novel multi-task vision-language framework (DisasterNet-Bangla) utilizing a shared Vision Transformer (ViT) encoder and a BanglaBERT decoder to simultaneously execute real-time flood damage severity classification and context-aware Bengali caption generation, employing dynamic loss balancing to mitigate task interference.
- **RO2 (Optimization & Efficiency Evaluation):** To implement parameter-efficient fine-tuning via Low-Rank Adaptation (LoRA) for adapting massive vision-language models on imbalanced, low-resource crisis datasets, and to critically evaluate its computational efficiency, memory footprint, and classification accuracy against traditional full fine-tuning and existing state-of-the-art unimodal baselines (e.g., BanglaMM-Disaster).
- **RO3 (Semantic Validation & Explainability):** To validate the semantic accuracy and explainability of the generated Bengali captions derived from noisy, ground-level social media imagery. This will be quantified using standard automated generation metrics (BLEU, ROUGE, METEOR) alongside comprehensive human-evaluation benchmarks to ensure practical utility in humanitarian disaster response.

Paper 1: Damage analysis via bidirectional multi-task cascaded multimodal fusion

- **1. Core Focus:** The study focuses on executing comprehensive multimodal damage analysis from a multi-task learning perspective, explicitly exploiting the intrinsic interdependencies among cascading subtasks, such as informativeness prediction, humanitarian categorization, and severity assessment.
- **2. Architecture Used:** The authors propose a Bidirectional Multi-Task Cascaded Multimodal Fusion (BiMCF) framework. It utilizes pre-trained BERT and Vision Transformer (ViT) as language and image backbones, incorporating Multimodal Task Cascading (MTC) gates (forward and backward flows) to enable task-specific feature extraction and fusion.
- **3. Scale & Dataset:** The model is evaluated on the standard CrisisMMD benchmark dataset, which consists of multimodal social media posts (image-text pairs) collected during seven major natural disasters.
- **4. Key Achievement:** The proposed BiMCF approach achieved state-of-the-art results across multiple baselines. By encoding task interdependencies directly into the model structure, it successfully outperformed existing single-task and parallel multi-task methods, achieving up to 93.4% accuracy on the severity prediction task.
- **5. Limitations/Gap:** While existing multi-task methods rely on shared multimodal representations, they fail to explicitly explore the dynamic, fine-grained relationships between cascaded tasks. Furthermore, this study is constrained to English datasets, leaving a gap in low-resource linguistic contexts.
- **6. DisasterNet-Bangla Integration:** This research provides a direct architectural foundation for the multi-task learning (MTL) framework of DisasterNet-Bangla. It validates the strategy of utilizing a shared encoder for simultaneous damage classification (Task Head 1) and caption generation (Task Head 2).
- **7. Thesis Placement:** This paper is highly relevant for Chapter 2 (Literature Review: Multimodal Multi-Task Learning) and Chapter 3 (Methodology: Multi-Task Integration).
- **8. Actionable Takeaway:** Implement a feature-sharing or cascaded strategy between the classification and captioning heads to ensure they mutually enhance the extraction of relevant visual features, maximizing computational efficiency and model accuracy.

Paper 2: Explainable flood damage assessment using multi-atrous self-attention and vision-language integration

- **1. Core Focus:** The paper introduces a multimodal framework that integrates semantic segmentation with explainable artificial intelligence (XAI) to detect flood-affected infrastructure and generate interpretable, natural language disaster descriptions.
- **2. Architecture Used:** The framework utilizes a Multi-Atrous Self-Attention (MASA) mechanism (extending DeepLabV3+ with a truncated ResNet-50 backbone) for segmentation, coupled with a Bootstrapped Language Image Pre-training (BLIP-2) vision-language model enhanced via Low-Rank Adaptation (LoRA) for textual generation.

- **3. Scale & Dataset:** The study leverages two distinct datasets: FloodNet, comprising high-resolution UAV images for segmentation, and LADI-v2, containing 10,000 image-caption pairs for vision-language modeling.
- **4. Key Achievement:** The MASA model achieved a mean Intersection over Union (mIoU) of 73.78% on FloodNet, while the LoRA-fine-tuned BLIP-2 model achieved a BLEU score of 80.77%, demonstrating an effective synergy between quantitative spatial detection and fluent linguistic reporting.
- **5. Limitations/Gap:** The methodology relies on separate datasets for segmentation and captioning, and the linguistic component is exclusively tailored to English, highlighting a significant novelty gap for unified frameworks addressing low-resource languages like Bengali.
- **6. DisasterNet-Bangla Integration:** This paper validates the use of LoRA for parameter-efficient fine-tuning in disaster scenarios. It justifies the novelty of DisasterNet-Bangla, which aims to bridge the identified gap by performing joint damage assessment and Bengali captioning.
- **7. Thesis Placement:** This study should be cited in Chapter 2 (Literature Review: Vision-Language Integration and Explainability in Disasters).
- **8. Actionable Takeaway:** Utilize LoRA to fine-tune the massive parameters of the captioning decoder (BanglaBERT/LSTM) and the shared Vision Encoder, ensuring competitive performance in resource-constrained environments typical of Bangladesh.

Paper 3: ReliefNet: A novel and reliable multimodal framework for real-time disaster severity classification

- **1. Core Focus:** The research presents ReliefNet, a real-time, explainable multimodal framework designed to mitigate modality imbalance and noisy social media content for robust disaster severity assessment.
- **2. Architecture Used:** The system integrates DualEmbedNet (a dual-encoder transformer for text) and DisasterNet (a CNN with channel-wise attention for images). These are merged using an accuracy-weighted decision-level fusion mechanism, complemented by CLIP embeddings and post-hoc XAI tools (Grad-CAM, LIME).
- **3. Scale & Dataset:** The authors constructed a massive, unified dataset of 14,996 image-text pairs by amalgamating CrisisMMD with the Turkey-Syria Earthquake Dataset (TSEqD), utilizing unsupervised K-Means clustering for pseudo-labeling.
- **4. Key Achievement:** ReliefNet achieved an exceptional F1-score of 97.00% under multimodal settings, demonstrating superior robustness, stability under missing-modality conditions, and real-time operational efficiency on a single GPU.
- **5. Limitations/Gap:** The decision-level fusion approach lacks fine-grained cross-modal interaction modeling. Additionally, the dataset is restricted to English, overlooking the critical need for disaster assessment tools in low-resource linguistic regions.
- **6. DisasterNet-Bangla Integration:** Provides a rigorous benchmark for the damage assessment head (Task Head 1). The methodology for unifying and processing CrisisMMD data offers a procedural template for creating the custom Bengali dataset.
- **7. Thesis Placement:** Relevant for Chapter 2 (Literature Review: Multimodal Disaster Severity Assessment) and Chapter 4 (Experiments & Results: Baseline Comparisons).

- **8. Actionable Takeaway:** Adopt comprehensive evaluation metrics (Accuracy, Precision, Recall, F1-Score) for the classification head, and ensure the data preprocessing pipeline robustly handles noisy textual translations to maintain label integrity.

Paper 4: CrisisViT: A Robust Vision Transformer for Crisis Image Classification

- **1. Core Focus:** The study investigates the application of state-of-the-art Vision Transformer (ViT) architectures for the automatic classification and categorization of crisis imagery sourced from social media.
- **2. Architecture Used:** The authors developed CrisisViT, a Transformer-based deep neural network adapted from the standard ViT, configured to process image patches via self-attention mechanisms.
- **3. Scale & Dataset:** The model was pre-trained on the extensive Incidents1M dataset (comprising over 1 million disaster-related images) and fine-tuned/evaluated on the Crisis Image Benchmark (which aggregates datasets like CrisisMMD) for tasks including damage severity estimation and disaster type classification.
- **4. Key Achievement:** The research established that Transformer-based architectures significantly outperform traditional Convolutional Neural Networks (CNNs) in crisis image classification. Pre-training on domain-specific data (Incidents1M) yielded an absolute performance gain of 1.25% across four distinct evaluation tasks.
- **5. Limitations/Gap:** The study is strictly unimodal, focusing solely on visual feature extraction and classification, and completely omits natural language generation or multimodal cross-lingual capabilities.
- **6. DisasterNet-Bangla Integration:** This paper serves as the primary architectural justification for selecting a Vision Transformer (ViT) as the Shared Vision Encoder in the DisasterNet-Bangla framework, rather than relying exclusively on traditional CNNs.
- **7. Thesis Placement:** Highly applicable to Chapter 3 (Methodology: Shared Vision Encoder Design) and Chapter 2 (Literature Review: Vision Transformers).
- **8. Actionable Takeaway:** Implement a pre-trained ViT as the core feature extractor to capture rich, global visual contexts from flood images, establishing a robust foundation that will simultaneously feed both the damage classifier and the BanglaBERT captioning decoder.

Paper 5: Deep Learning Methods of Satellite Image Processing for Monitoring of Flood Dynamics in the Ganges Delta, Bangladesh

- **1. Core Focus:** The paper explores the application of advanced Deep Learning (DL) and Geographic Information System (GIS) tools to process satellite imagery for the real-time monitoring of flood dynamics and landscape changes specifically within the Ganges River Delta in Bangladesh.
- **2. Architecture Used:** The methodology utilizes a Multilayer Perceptron (MLP) Classifier algorithm embedded within the GRASS GIS scripting framework to automatically detect and categorize flooded land cover types.

- **3. Scale & Dataset:** The analysis is conducted using a time series of multispectral Landsat 8-9 OLI/TIRS satellite images, captured during pre- and post-flood periods (March and November) across the years 2014, 2021, 2022, and 2023.
- **4. Key Achievement:** The study successfully mapped and quantified the extent of inundated areas and subsequent landscape recovery, proving that DL-based satellite image classification can highly automate and improve the accuracy of regional hazard assessments.
- **5. Limitations/Gap:** The reliance on top-down satellite imagery lacks the immediate, ground-level, human-centric perspective necessary for rapid, localized damage assessment and immediate humanitarian deployment, which social media imagery provides.
- **6. DisasterNet-Bangla Integration:** This paper is crucial for contextualizing the socio-economic and geographical motivation of the thesis. It justifies the urgent need for localized, rapid-response AI systems specifically tailored to the flood-prone environment of Bangladesh.
- **7. Thesis Placement:** Central to Chapter 1 (Introduction: Motivation, Background, and Real-World Impact).
- **8. Actionable Takeaway:** Use the findings from this paper in the introduction to articulate the specific vulnerabilities of Bangladesh to flooding. Emphasize how DisasterNet-Bangla's ground-level, rapid-response framework directly addresses the latency and language barriers inherent in existing top-down, English-centric monitoring systems.

Paper 6: BiCap: Bangla Image Captioning Using Attention-based Encoder-Decoder Architecture

- **1. Core Focus:** The study introduces an attention-based encoder-decoder framework designed to generate fluent and semantically grounded Bengali captions for images, explicitly addressing the dual challenges of semantic alignment and syntactic fluency in a low-resource language.
- **2. Architecture Used:** The model employs a truncated ResNet-50 as the visual feature encoder, integrated with a Bahdanau attention mechanism for dynamic spatial region selection, and a single-layer Long Short-Term Memory (LSTM) network as the sequential text decoder.
- **3. Scale & Dataset:** The architecture was trained and evaluated on the Chitron dataset, which is the largest publicly available Bengali image captioning corpus, comprising 15,438 human-annotated image-caption pairs scaled to 299×299 pixels.
- **4. Key Achievement:** The proposed BiCap framework significantly outperformed existing baselines, achieving superior quantitative metrics (BLEU, METEOR, ROUGE, and CIDEr) while demonstrating marked qualitative improvements in human-evaluated fluency, relevance, and semantic adequacy.
- **5. Limitations/Gap:** The framework is primarily evaluated on general-domain imagery, lacking domain-specific adaptation and vocabulary formulation necessary for highly specialized, catastrophic contexts such as flood damage assessment.

- **6. DisasterNet-Bangla Integration:** This research serves as a direct architectural reference for the caption generation component (Task Head 2) of DisasterNet-Bangla, validating the feasibility and effectiveness of coupling a CNN encoder with an attention-augmented RNN/LSTM decoder for Bengali.
- **7. Thesis Placement:** Highly relevant for Chapter 2 (Literature Review: Bengali Image Captioning) and Chapter 3 (Methodology: Captioning Head Design).
- **8. Actionable Takeaway:** Incorporate an explicit attention mechanism between the Shared Vision Encoder and the Bengali text decoder to ensure the model dynamically focuses on critical, damaged visual regions when generating context-aware disaster descriptors.

Paper 7: Image captioning in Bengali language using visual attention

- **1. Core Focus:** The research focuses on developing an end-to-end architecture to generate contextually relevant Bengali image descriptions by evaluating various combinations of Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) equipped with visual attention, supplemented by Explainable AI (XAI).
- **2. Architecture Used:** The study systematically compared multiple CNN encoders (VGG19, ResNet-50 v2, Inception v3) paired with various RNN decoders (GRU, LSTM, BiLSTM), augmented by a weighted-sum attention layer and analyzed via SHAP (Shapley Additive Explanations).
- **3. Scale & Dataset:** Experiments were conducted using a manually translated and grammatically corrected Bengali version of the standard Flickr 8k Dataset, comprising 8,000 images with 5 captions per image.
- **4. Key Achievement:** The configuration pairing VGG19 with an attention-augmented LSTM achieved peak performance, yielding a BLEU-4 score of 0.432 (43%), METEOR of 0.389 (39%), and ROUGE of 0.470 (47%). SHAP analysis successfully mapped pixel-level contributions to specific generated Bengali words.
- **5. Limitations/Gap:** The dataset is constrained to general, everyday scenarios, and the models exhibit difficulties in handling complex sentence structures and capturing domain-specific nuances, highlighting a need for context-aware training in specialized fields like disaster response.
- **6. DisasterNet-Bangla Integration:** Provides robust comparative baseline metrics for the captioning head. Furthermore, the integration of SHAP establishes a methodological pathway for rendering DisasterNet-Bangla's multimodal outputs interpretable to humanitarian stakeholders.
- **7. Thesis Placement:** Applicable to Chapter 2 (Literature Review: Attention Mechanisms in Bengali NLP) and Chapter 4 (Experiments & Results: Baseline Comparisons and Interpretability).
- **8. Actionable Takeaway:** Adopt beam search decoding (with a beam width of 3) during the inference phase of the captioning head to normalize sentence length and optimize the fluency of the generated Bengali text.

Paper 8: Transfer Learning for Low-Resource Bengali Image Captioning

- **1. Core Focus:** The paper investigates the application of transfer learning techniques to overcome critical data scarcity in low-resource Bengali image captioning by leveraging large-scale, pre-trained visual and language models.
- **2. Architecture Used:** The methodology utilizes pre-trained CNNs (such as ResNet or EfficientNet) for visual feature extraction, coupled with pre-trained language or sequence models for the decoder, employing selective layer freezing and fine-tuning strategies.
- **3. Scale & Dataset:** The approach targets limited Bengali image-caption datasets, aggressively supplemented by data augmentation techniques including image transformations and caption paraphrasing to artificially expand the training corpus.
- **4. Key Achievement:** The study demonstrated that initializing models with pre-trained weights and utilizing partial fine-tuning significantly enhances caption quality, fluency, and grammatical correctness compared to models trained entirely from scratch under low-resource constraints.
- **5. Limitations/Gap:** The research presents a generalized transfer learning approach without addressing the complexities of multi-task environments or the specialized lexicon required for crisis and disaster management.
- **6. DisasterNet-Bangla Integration:** Conceptually justifies DisasterNet-Bangla's core strategy of utilizing pre-trained foundation models (e.g., ViT and BanglaBERT) to mitigate the absence of massive, native Bengali disaster datasets.
- **7. Thesis Placement:** Central to Chapter 2 (Literature Review: Transfer Learning in Low-Resource Languages) and Chapter 3 (Methodology: Optimization Strategy).
- **8. Actionable Takeaway:** Apply rigorous data augmentation pipelines to the translated CrisisMMD dataset and utilize parameter-efficient fine-tuning (such as LoRA) to prevent overfitting while adapting massive pre-trained weights to the specialized disaster domain.

Paper 9: SARChat-Bench-2M: A Multi-Task Vision-Language Benchmark for SAR Image Interpretation

- **1. Core Focus:** The study establishes SARChat-Bench-2M, a comprehensive multi-task vision-language benchmark designed to facilitate deep semantic parsing, reasoning, and interpretation of Synthetic Aperture Radar (SAR) imagery.
- **2. Architecture Used:** The research evaluated 16 state-of-the-art Vision-Language Models (VLMs) of varying parameter scales (e.g., InternVL2.5, DeepSeekVL, Qwen2-VL), uniformly fine-tuned using Low-Rank Adaptation (LoRA) across all linear layers.
- **3. Scale & Dataset:** The authors constructed the SARChat-2M dataset containing approximately 2 million high-quality SAR image-text pairs, structured to support six distinct interpretive tasks including classification, fine-grained description, and spatial grounding.
- **4. Key Achievement:** The study proved that domain-specific instruction tuning enables edge-side VLMs ($\leq 5B$ parameters) to achieve remarkable multi-task capabilities, successfully bridging the semantic-visual gap to simultaneously perform structured classification and descriptive generation.

- **5. Limitations/Gap:** The benchmark is strictly confined to SAR imagery and English instructions, leaving a significant research void for optical disaster imagery interpretation in low-resource languages like Bengali.
- **6. DisasterNet-Bangla Integration:** Validates the efficacy of the multi-task learning (MTL) paradigm where a single unified model handles both discrete classification and generative descriptions. It also provides a robust template for formulating task-specific instruction prompts.
- **7. Thesis Placement:** Highly relevant for Chapter 2 (Literature Review: Multi-Task Vision-Language Models) and Chapter 3 (Methodology: Multi-Task Prompting and Integration).
- **8. Actionable Takeaway:** Utilize explicit task-specific prompts or identifiers (e.g., `[classify]`, `[caption]`) during the multi-task training phase to explicitly guide the shared encoder's attention, effectively mitigating task interference.

Paper 10: Co-Training Vision-Language Models for Remote Sensing Multi-Task Learning

- **1. Core Focus:** The paper proposes RSCoVLM, a unified multi-task vision-language foundation model designed to simultaneously handle diverse remote sensing (RS) tasks via cooperative training (co-training) and a dynamic resolution strategy.
- **2. Architecture Used:** A generative VLM paradigm employing a Vision Transformer (ViT) encoder, a cross-modal alignment projection, and a unified Large Language Model (LLM) decoder (Qwen2.5-VL), optimized via next-token prediction.
- **3. Scale & Dataset:** The model was trained on a meticulously curated dataset of 1.55 million images and 1.46 million conversations, integrating multiple RS datasets and incorporating a novel "Zoom-in Chain" strategy for ultra-high-resolution imagery.
- **4. Key Achievement:** RSCoVLM achieved state-of-the-art performance across multiple RS benchmarks (including scene classification and visual question answering), rivaling specialized single-task expert models by employing dynamic sampling weights during joint training.
- **5. Limitations/Gap:** The authors acknowledge persistent challenges with task interference in multi-task joint training, observing that performance gains in one task occasionally occur at the expense of another.
- **6. DisasterNet-Bangla Integration:** Strongly supports the co-training methodology of DisasterNet-Bangla. It provides critical insights into balancing multi-task learning to prevent catastrophic forgetting and optimizing joint classification and captioning objectives.
- **7. Thesis Placement:** Applicable to Chapter 2 (Literature Review: Co-Training and Multi-Task Remote Sensing) and Chapter 3 (Methodology: Multi-Task Optimization and Loss Balancing).
- **8. Actionable Takeaway:** Implement a dynamic, weighted loss function during the joint training of Task Head 1 and Task Head 2. Start with uniform sampling ratios and iteratively adjust the weights to increase the learning proportion of the underperforming task, thereby minimizing inter-task competition.

Paper 11: PetroMind: A multimodal petrographic model for rock image classification and lithological description generation

- **1. Core Focus:** The study focuses on jointly addressing complex image classification and detailed lithological description generation by developing a domain-specific multimodal foundation model.
- **2. Architecture Used:** The proposed PetroMind framework incorporates a Vision Transformer (ViT) based vision encoder, a CLIP-based vision-text alignment module, and a Qwen2.5-VL-based Large Language Model (LLM) decoder, all optimized via Low-Rank Adaptation (LoRA).
- **3. Scale & Dataset:** The model utilizes the ROCK-13 dataset, which comprises 5,845 manually annotated images across 13 classes, programmatically expanded into 58,450 classification instructions and 272 specialized description generation instructions.
- **4. Key Achievement:** PetroMind achieved a Macro-F1 score of 97.35% for classification and successfully established a novel LLM-based metric (SA-Rock) for text generation, demonstrating robust few-shot and long-tail image learning capabilities.
- **5. Limitations/Gap:** The framework is evaluated strictly within the geological domain of plutonic intrusive rocks, and the integration of multi-task parameter sharing revealed challenges in simultaneously balancing high-resource classification and low-resource generative tasks.
- **6. DisasterNet-Bangla Integration:** This research provides a robust architectural template for executing simultaneous visual classification and text generation using a shared vision encoder and an autoregressive decoder optimized by parameter-efficient fine-tuning.
- **7. Thesis Placement:** Highly relevant for Chapter 2 (Literature Review: Multimodal Foundation Models) and Chapter 3 (Methodology: Multi-Task Architecture).
- **8. Actionable Takeaway:** Investigate the strategic use of task-specific LoRA adapters versus a single shared LoRA adapter to effectively balance the loss gradients between the data-rich damage classification head and the data-scarce Bengali captioning head.

Paper 12: BanglaMM-Disaster: A Multimodal Transformer-Based Deep Learning Framework for Multiclass Disaster Classification in Bangla

- **1. Core Focus:** The research presents a multimodal deep learning framework specifically engineered for multiclass disaster classification in the Bengali language, leveraging both visual and textual data extracted from social media.
- **2. Architecture Used:** The study employs an early fusion framework integrating a ResNet50 visual encoder with a pre-trained text encoder (such as BanglaBERT, mBERT, or XLM-RoBERTa), combined via cross-attention mechanisms and dense classification layers.
- **3. Scale & Dataset:** Evaluations were conducted on a custom multimodal dataset processed by translating English and Banglish disaster texts into Bengali, formatted to 224x224 image resolutions, and encompassing 9 distinct disaster categories.

- **4. Key Achievement:** The multimodal fusion approach utilizing mBERT and ResNet50 achieved a peak F1-score of 83.76%, significantly outperforming unimodal text-only and image-only baselines within the Bengali linguistic context.
- **5. Limitations/Gap:** The framework is strictly limited to discriminative categorization (Task Head 1) and entirely lacks the generative capacity required to produce descriptive Bengali disaster captions (Task Head 2).
- **6. DisasterNet-Bangla Integration:** This paper serves as the primary regional baseline for the damage classification component, directly validating the architectural feasibility of pairing ResNet or ViT with BanglaBERT for disaster-specific applications in Bangladesh.
- **7. Thesis Placement:** Central to Chapter 2 (Literature Review: Multimodal Disaster Classification in Bengali) and Chapter 4 (Experiments & Results: Baseline Comparisons).
- **8. Actionable Takeaway:** Utilize the reported F1-score of 83.76% as a quantitative baseline to exceed, and adopt their rigorous text preprocessing pipeline to effectively handle and normalize noisy Banglish-to-Bengali translations.

Paper 13: AI and NLP in Social Good: Enhancing Emergency Response and Crisis Handling via Text Analytics

- **1. Core Focus:** The paper examines the application of Artificial Intelligence (AI) and Natural Language Processing (NLP) for social good, specifically focusing on enhancing real-time emergency response and crisis management through unstructured text analytics.
- **2. Architecture Used:** A conceptual and analytical review framework that explores various NLP methodologies, including sentiment analysis, text classification, Named Entity Recognition (NER), and topic modeling utilizing advanced models like BERT and GPT.
- **3. Scale & Dataset:** A comprehensive review study analyzing heterogeneous textual data sources (social media, emergency calls, news reports), noting that social media networks like Twitter and Facebook constitute 40% of actionable text data during the crisis response phase.
- **4. Key Achievement:** The study synthesized a cohesive pipeline demonstrating how NLP automates the extraction of life-saving insights, directly optimizing resource allocation, situational awareness, and humanitarian logistics.
- **5. Limitations/Gap:** The research is heavily skewed towards English-centric text analytics, leaving a substantial gap regarding the integration of visual modalities (multimodal systems) and the deployment of such technologies in low-resource languages.
- **6. DisasterNet-Bangla Integration:** Provides the socio-technical and humanitarian justification for the thesis, framing the generation of localized Bengali disaster captions as a critical intervention for overcoming language barriers in rapid humanitarian response.
- **7. Thesis Placement:** Crucial for Chapter 1 (Introduction: Motivation and Social Impact) and Chapter 2 (Literature Review: AI for Humanitarian Aid).

- **8. Actionable Takeaway:** Structure the thesis narrative to emphasize how DisasterNet-Bangla reduces humanitarian information overload by automating ground-level situational awareness directly in the native language of the affected populace.

Paper 14: LoRA: Low-Rank Adaptation of Large Language Models

- **1. Core Focus:** The study proposes Low-Rank Adaptation (LoRA) to efficiently fine-tune massive pre-trained language and foundation models by significantly reducing the number of trainable parameters while maintaining target model quality.
- **2. Architecture Used:** The method modifies standard Transformer architectures by freezing pre-trained model weights and injecting trainable low-rank decomposition matrices (A and B) parallel to existing dense layers, specifically targeting the query (W_q) and value (W_v) projection matrices within self-attention modules.
- **3. Scale & Dataset:** Extensively benchmarked on massive models including GPT-3 (175 Billion parameters) and GPT-2 across diverse datasets such as WikiSQL, MultiNLI, and the E2E NLG Challenge.
- **4. Key Achievement:** LoRA reduced the number of trainable parameters by 10,000 times and computational hardware requirements by 3 times for GPT-3, matching or exceeding full fine-tuning performance with zero additional inference latency.
- **5. Limitations/Gap:** The original methodology was explicitly tailored for and evaluated on NLP tasks with massive LLMs, requiring subsequent empirical validation for cross-modal (vision-language) adaptation on mid-sized encoders.
- **6. DisasterNet-Bangla Integration:** Forms the foundational theoretical basis for the optimization strategy of the thesis, justifying how the framework will successfully fine-tune parameter-heavy models (ViT and BanglaBERT) within the resource-constrained hardware environments typical of Bangladesh.
- **7. Thesis Placement:** Central to Chapter 3 (Methodology: Optimization and Fine-Tuning Strategy).
- **8. Actionable Takeaway:** Apply LoRA specifically to the query and value projection matrices within the attention blocks of both the Shared Vision Encoder (ViT) and the Bengali text decoder (BanglaBERT), utilizing a low rank (e.g., $r=4$ or $r=8$) to maximize computational efficiency without sacrificing classification or generative accuracy.

Paper 15: CrisisMMD: Multimodal Twitter Datasets from Natural Disasters

- **1. Core Focus:** The paper introduces a large-scale, human-annotated multimodal dataset composed of paired text and imagery collected from Twitter to facilitate advanced computational processing and situational awareness during natural disasters.
- **2. Architecture Used:** A dataset construction methodology involving programmatic data filtering, deduplication, and a multi-tiered crowdsourced annotation pipeline utilizing the Figure Eight platform to establish rigorous ground-truth labels.

- **3. Scale & Dataset:** The corpus comprises thousands of annotated image-text pairs (sampled from millions of raw tweets) spanning seven major 2017 natural disasters, including Hurricane Harvey, Hurricane Irma, and the Sri Lanka floods.
- **4. Key Achievement:** Provided the first comprehensive public multimodal benchmark for disaster response, successfully standardizing critical humanitarian tasks: informativeness prediction, humanitarian categorization, and damage severity assessment.
- **5. Limitations/Gap:** The dataset is restricted entirely to the English language and exhibits significant class imbalances (e.g., a disproportionate ratio between severe damage and little/no damage instances), limiting its direct deployment in non-English, low-resource operational regions.
- **6. DisasterNet-Bangla Integration:** This represents the exact foundational source dataset that will be programmatically translated, manually validated, and architecturally adapted to construct the custom Bengali multimodal dataset for the thesis.
- **7. Thesis Placement:** Highly relevant for Chapter 1 (Introduction: Data Scarcity) and Chapter 3 (Methodology: Dataset Creation and Preprocessing).
- **8. Actionable Takeaway:** Adopt the exact label taxonomy from this dataset for Task Head 1 (Damage Severity: Severe, Mild, Little/No Damage) to ensure the Bengali adaptation remains comparable to international benchmarks, while proactively implementing data augmentation techniques to address the inherent severity class imbalance during training.

Paper 16: BanglaBERT: Language Model Pretraining and Benchmarks for Low-Resource Language Understanding Evaluation in Bangla

- **1. Core Focus:** The research focuses on pretraining a BERT-based Natural Language Understanding (NLU) model specifically tailored for the Bengali language, named BanglaBERT, and establishing the comprehensive Bangla Language Understanding Benchmark (BLUB) to evaluate models in a low-resource linguistic context.
- **2. Architecture Used:** The model employs an ELECTRA-based pretraining objective known as Replaced Token Detection (RTD), which jointly trains a generator and a discriminator network. It utilizes a custom WordPiece vocabulary of 32,000 subword tokens to capture morphological nuances and code-switching.
- **3. Scale & Dataset:** BanglaBERT was pretrained on "Bangla2B+", a meticulously curated 27.5 GB dataset comprising 2.18 billion tokens extracted from 110 popular Bengali websites. It was evaluated on four distinct downstream tasks (sentiment classification, natural language inference, named entity recognition, and question answering).
- **4. Key Achievement:** In supervised fine-tuning setups, BanglaBERT achieved state-of-the-art performance across all BLUB benchmark tasks, significantly outperforming existing multilingual models (such as mBERT and XLM-R) while demonstrating superior compute, memory, and sample efficiency.
- **5. Limitations/Gap:** The study is primarily concentrated on Natural Language Understanding (NLU) and leaves the exploration of Natural Language Generation (NLG)

tasks and multimodal model initialization as future work. Additionally, despite filtering, the pretraining corpus may still contain latent noisy or objectionable content.

- **6. DisasterNet-Bangla Integration:** This study provides the exact foundational architecture and pre-trained checkpoints for the Bengali caption generation component (Task Head 2) of the thesis, ensuring the model possesses robust, context-aware Bengali linguistic capabilities without requiring ground-up pretraining.
- **7. Thesis Placement:** Central to Chapter 2 (Literature Review: Foundational Bengali NLP Models) and Chapter 3 (Methodology: Captioning Decoder Architecture).
- **8. Actionable Takeaway:** Initialize the captioning decoder of DisasterNet-Bangla with pre-trained BanglaBERT weights to leverage its deep semantic understanding of Bengali, and apply parameter-efficient fine-tuning (LoRA) to adapt it specifically to the specialized disaster management lexicon.

Paper 17: An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale

- **1. Core Focus:** The study investigates the direct application of Transformer architectures to image recognition, demonstrating that reliance on Convolutional Neural Networks (CNNs) is unnecessary and that pure self-attention models can achieve exceptional performance at scale.
- **2. Architecture Used:** The authors developed the Vision Transformer (ViT), which reshapes 2D images into a 1D sequence of flattened, fixed-size patches. These patches are mapped to constant latent vectors, combined with positional embeddings, and processed through a standard Transformer encoder consisting of alternating multi-head self-attention and MLP blocks.
- **3. Scale & Dataset:** ViT models of varying capacities (Base, Large, Huge) were pre-trained on massive datasets, including ImageNet-21k (14 million images) and JFT-300M (300 million images), and subsequently fine-tuned on mid-sized benchmarks like ImageNet, CIFAR-100, and VTAB.
- **4. Key Achievement:** When pre-trained on sufficiently large datasets, ViT models attained state-of-the-art results, matching or exceeding highly optimized CNNs (e.g., ResNet-based Big Transfer) while requiring substantially fewer computational resources to pre-train.
- **5. Limitations/Gap:** ViT possesses significantly less image-specific inductive bias (such as translation equivariance and two-dimensional neighborhood structure) compared to CNNs. Consequently, it underperforms when trained from scratch on insufficient amounts of data, requiring massive pre-training corpora to generalize effectively.
- **6. DisasterNet-Bangla Integration:** This paper establishes the theoretical and structural justification for selecting a Vision Transformer (ViT) as the Shared Vision Encoder in the DisasterNet-Bangla framework.
- **7. Thesis Placement:** Highly applicable to Chapter 2 (Literature Review: Vision Transformers and Foundation Models) and Chapter 3 (Methodology: Shared Vision Encoder Design).
- **8. Actionable Takeaway:** Utilize a ViT model previously pre-trained on a massive dataset (e.g., ImageNet-21k) as the shared visual feature extractor to bypass its

data-hungry nature, feeding its rich, global patch embeddings directly into both the damage classifier and the BanglaBERT captioning decoder.